LESSON PLAN

EMI/EMC (14)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	41
Total No. of Lesson Plans	46	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Explanation of stability and control augmentation system, attitude base stabilization system.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Explanation of stability and control augmentation system, attitude base stabilization system.
5. Specific Objectives	14.2.1.1 Trainee will be able to explain the functions of AFCS.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of AFCS block diagrams and signal flow charts from IB; Instructor supervision required for diagram interpretation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of the AFCS?
(b) Define stability augmentation in the context of rotorcraft.
(c) List two types of control augmentation systems.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition of Stability Augmentation System • Purpose of reducing pilot workload • Basic components of SAS • Sensor inputs for SAS • Actuator response mechanisms	L & Ds	PPT, IB	15 min
(b)	• Attitude Based Stabilization System principles • Pitch and roll axis stabilization • Yaw rate damping functions • Integration with main autopilot • Fail-safe modes and reversion	L & Ds	PPT, PDF	20 min
(c)	• Comparison of SAS and ABS systems • Operational benefits of AFCS • Limitations of augmentation systems • Maintenance implications • Transition to next lesson topics	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how the AFCS improves helicopter handling qualities.
(b) Describe the difference between stability augmentation and attitude stabilization.
(c) Identify the primary sensors used in an attitude-based stabilization system.

12. Home Assignment and Follow Up	Prepare a brief note on the facilities available through the helicopter autopilot system as covered in LP 42, focusing on how it extends the capabilities of the AFCS.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Explanation of helicopter autopilot system and various facilities available through it.	Refer to the master syllabus reference list	LP 42

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Instructor, Flight Lieutenant, P No. 12345	Senior Instructor, Major, P No. 67890
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min